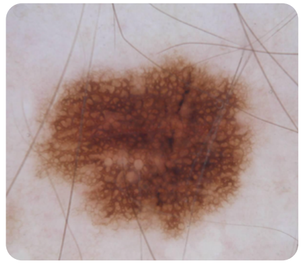
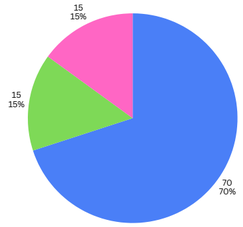
Dataset
ISIC 2019
(BCC,Melanoma,Nevus)



Preprocessing

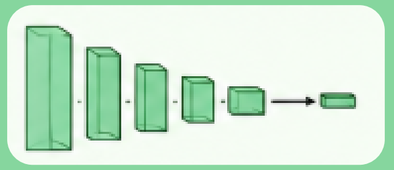
- Resize
- color normalization
- Data Augmentation
- (Flip,rotate,zoom,Shift,Brightness)
- To Tensor
- Normalization

Data Split



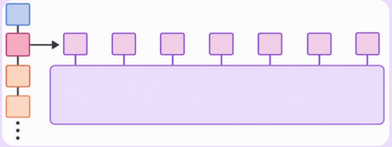
Training
Validation
Testing

Model1:MobileNetV2



Pre Trained on
ImageNet+Custom classifier
Head(3 classses)
Output Probabilities

Model2:Dinov2 (ViT)



Pre Trained on Large Scale
Data + Custom classifier
Head(3 classes)
Output Probabilities

Weighted Ensemble

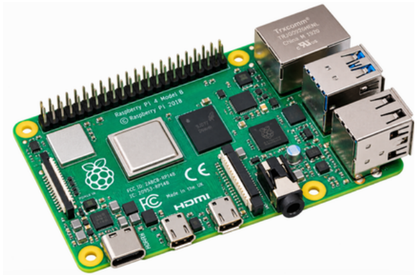
$$P_{ensemble} = w_M \cdot P_M + w_D \cdot P_D$$

where,

$$w_M + w_D = 1$$

(Best Weights Found:
$w_M = 0.4,\ w_D = 0.6$)

Edge Deployment on Raspberry Pi



Classification Result

Nevus:98.46%

